Name: Solutions ______ (please print)

Signature: ____________________

ECE 2202 – Quiz 2

February 18, ~~2025~~ 2026

1. Show all work necessary to complete the problem. Use additional sheets of paper as needed.

2. Show all units in solutions and figures. Units in the quiz will be included between square brackets.

3. You will have 40 minutes to work on the quiz.

4. Write your ID number on every page that you show your work.

__________/20

ID number: ______________________

Problem 1/1 (20 pts)

A Device can be modeled as an independent current source in parallel with a resistance. The Device is shown in Figure 1. The relationship between the voltage and current for the Device is given in the plot in Figure 2. Two identical versions of this Device are inserted in the circuit shown in Figure 3. The polarities of the Devices are indicated with the terminal labels 1 and 2.

a) Find a Norton equivalent circuit model for the Device. Draw the model, labeling the components with numerical values.

b) Find the Thevenin equivalent of the circuit in Figure 3 as seen by the dependent voltage source. Draw the Thevenin equivalent circuit with clear labels.

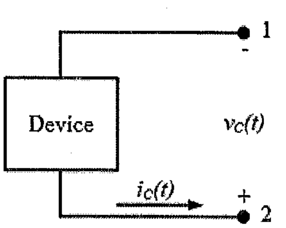

Figure 1

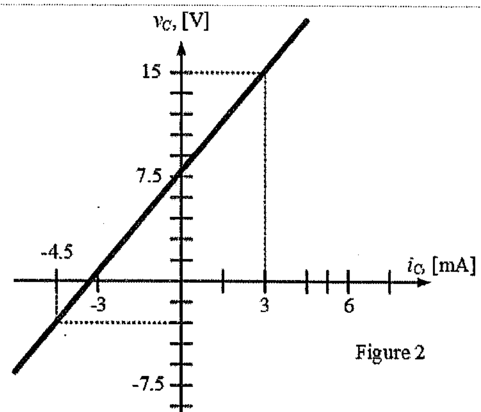

Figure 2

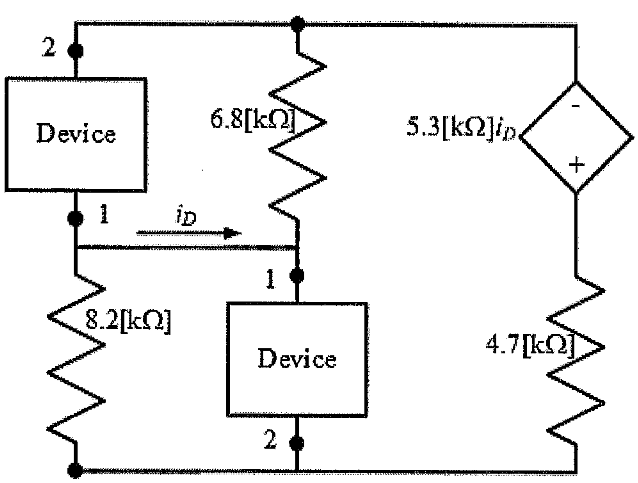

Figure 3

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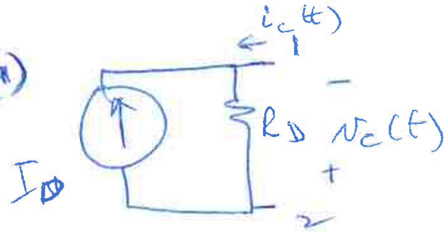
Room for extra work

(a)



<u>KCL at Node 1</u>

$$-I_D - \frac{v_c}{R_D} = i_c$$

① $(-4.5[mA], -3[V])$

$$-I_D + \frac{3[V]}{R_D} = -4.5[mA]$$

② $(3[mA], 15[V])$

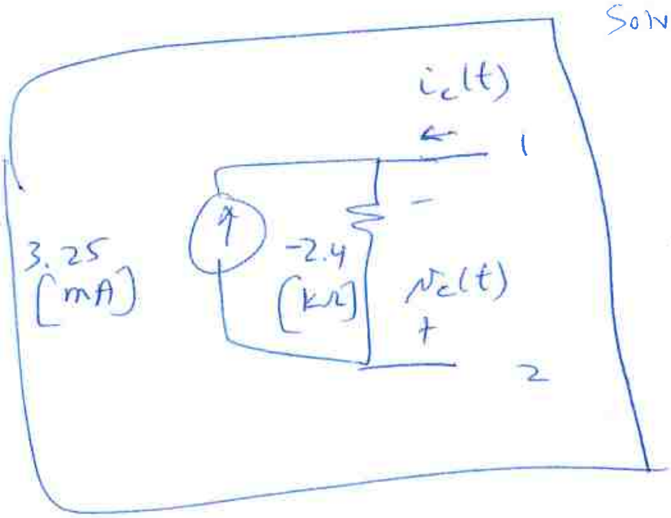
$$-I_D - \frac{15[V]}{R_D} = 3[mA]$$

Solve: $\frac{3}{R_D} + .0045 = \frac{-15}{R_D} - 0.003 = I_D$

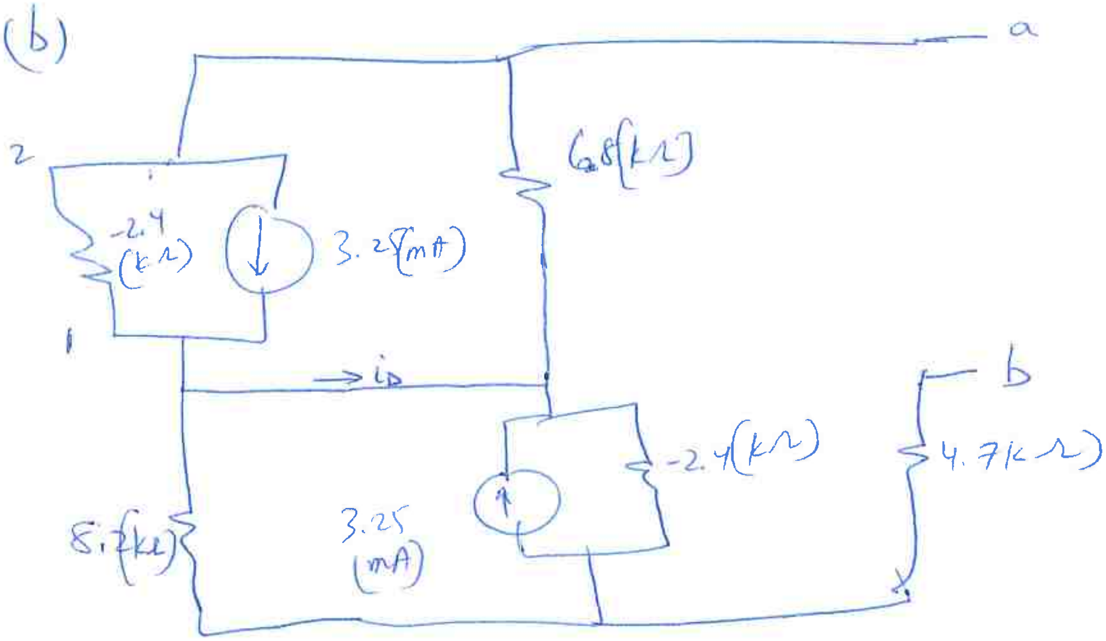
$$\frac{3+15}{R_D} = -.0075$$

$$\boxed{\begin{aligned} R_D &= -2.4[k\Omega] \\ I_D &= 3.25[mA] \end{aligned}}$$

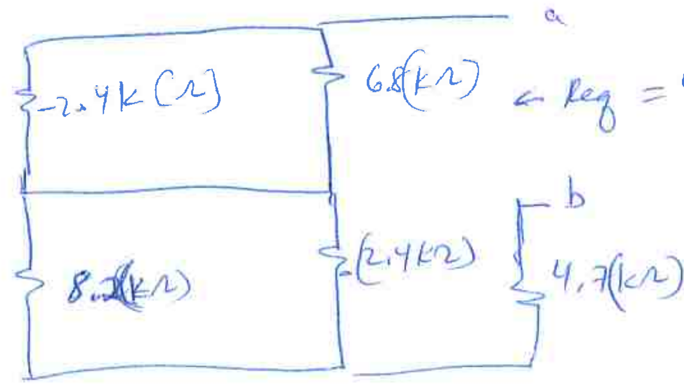
$\leftarrow$



(b)



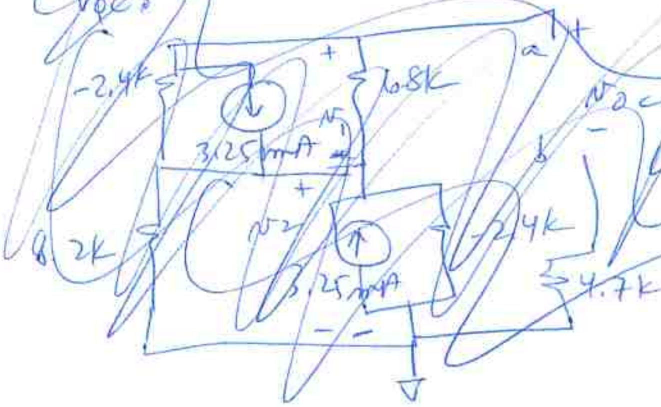
Req:



$$\leftarrow R_{eq} = 4.7[k\Omega] + (-2.4k) \,||\, ((6.8k) + (-2.4k) \,||\, (8.2k))$$

$$= -2.4022[k\Omega]$$

~~C Voc:~~



~~$\frac{v_1 - v_2}{-2.4k} + 3.25mA + \frac{v_1 - v_2}{6.8k}$~~

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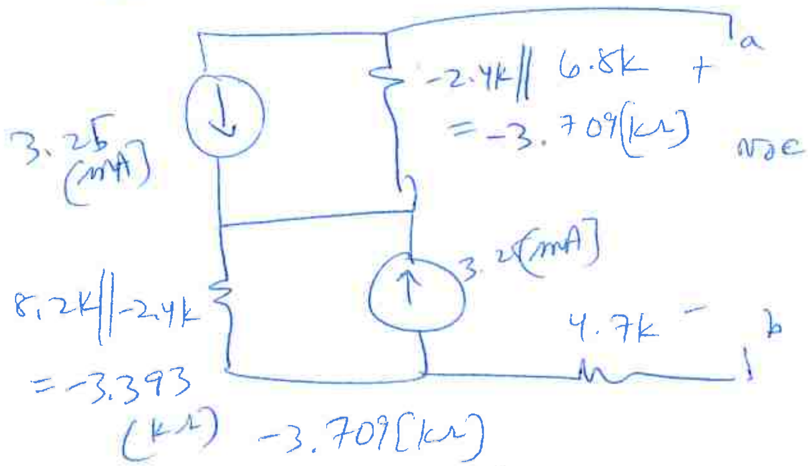
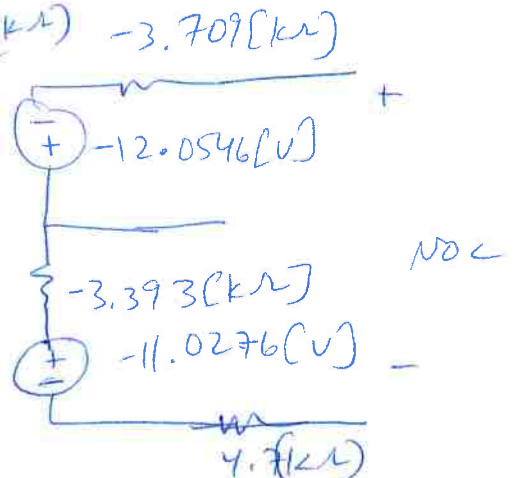
v_{oc}:



Source transformation →



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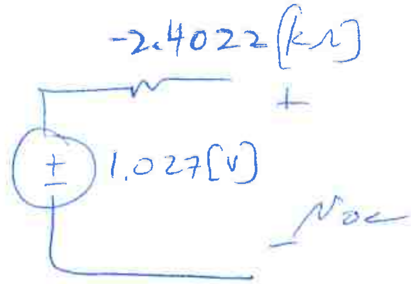
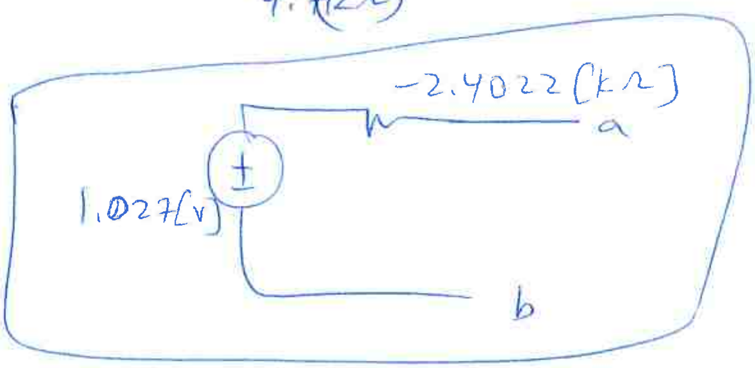


$\boxed{v_{oc} = 1.027(V)}$



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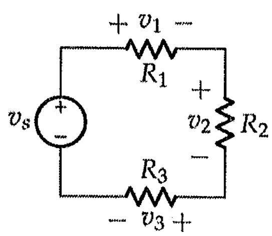
$$i = \frac{\mathrm{d}q}{\mathrm{d}t} \qquad v = \frac{\mathrm{d}w}{\mathrm{d}q} \qquad p = \frac{\mathrm{d}w}{\mathrm{d}t} = \frac{\mathrm{d}w}{\mathrm{d}q}\frac{\mathrm{d}q}{\mathrm{d}t} = iv \qquad i = C\frac{\mathrm{d}v}{\mathrm{d}t} \qquad v = L\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$v = iR \qquad R_1||R_2 = \frac{1}{\frac{1}{R_1}+\frac{1}{R_2}} = \frac{R_1R_2}{R_1+R_2} \qquad p = Cv\frac{\mathrm{d}v}{\mathrm{d}t} \qquad p = Li\frac{\mathrm{d}i}{\mathrm{d}t}$$

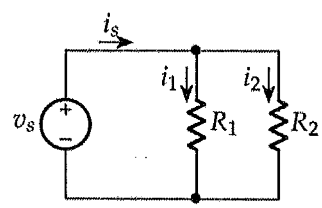
$$p = iv = i^2R = \frac{v^2}{R} \qquad w = \frac{1}{2}Li^2 \qquad w = \frac{1}{2}Cv^2$$

$$R = \frac{v}{i}$$

$$v_C(t) = v_C(\infty) + [v_C(T_0) - v_C(\infty)]\,e^{-(t-T_0)/\tau}$$
$$i_L(t) = i_L(\infty) + [i_L(T_0) - i_L(\infty)]\,e^{-(t-T_0)/\tau}$$
(Switch action at $t = T_0$)



$$v_1 = \frac{R_1}{R_1+R_2+R_3}v_s \qquad v_2 = \frac{R_2}{R_1+R_2+R_3}v_s \qquad v_3 = \frac{R_3}{R_1+R_2+R_3}v_s$$

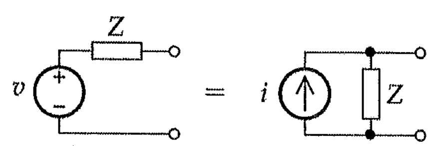


$$i_1 = \frac{R_2}{R_1+R_2}i_s \qquad i_2 = \frac{R_1}{R_1+R_2}i_s$$

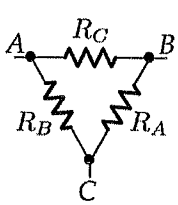
$$v_B = \underbrace{V_B}_{\text{DC}} + \underbrace{v_b}_{\text{AC}}$$

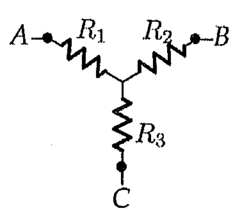
$\overline{V}_b \rightarrow$ Phasor notation

$$CMRR = 20\log_{10}\frac{|A_d|}{|A_{cm}|}$$



$$v = i \cdot Z \qquad i = \frac{v}{Z}$$





$$R_1 = \frac{R_BR_C}{R_A+R_B+R_C} \qquad R_A = \frac{R_1R_2+R_2R_3+R_1R_3}{R_1}$$
$$R_2 = \frac{R_AR_C}{R_A+R_B+R_C} \qquad R_B = \frac{R_1R_2+R_2R_3+R_1R_3}{R_2}$$
$$R_3 = \frac{R_AR_B}{R_A+R_B+R_C} \qquad R_C = \frac{R_1R_2+R_2R_3+R_1R_3}{R_3}$$